

COUCH-POTATO: DATA-DRIVEN MOVIE RECOMMENDATION BLUEPRINT

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Tagline: *Leveraging data intelligence to personalize movie experiences and boost audience engagement*

1. Overview

This project focuses on building a Movie Recommendation System using the MovieLens dataset, which contains user ratings, movie metadata, and linking information. The main goal of the project is to analyze user–movie interaction patterns and build a data-driven recommendation model that delivers personalized suggestions.

The workflow involved several critical steps: data collection, data cleaning and preparation, exploratory data analysis (EDA), feature engineering, model development, and finally deployment through an interactive Streamlit application.

The central objective was to enhance the movie discovery experience by recommending relevant films to users based on their previous ratings and preferences. Through careful analysis of user ratings, genre patterns, and rating distributions, the project successfully created a working recommendation model.

2. Problem Statement

Streaming platforms and large movie catalogs overwhelm users. Browsing is time-consuming and static “popular” lists do not respect individual taste. We face a twofold problem:

- How to infer each user’s taste reliably from sparse historical ratings
- How to serve recommendations in a usable, immediate format. Without good recommendations, platforms suffer reduced engagement, lower watch time, and churn.

3. Central Business Question

How can historical user ratings and movie metadata be used to generate accurate, personalized, and scalable movie recommendations that improve user discovery and engagement?

4. Business Objectives

1. **Predictive Objective:** Build a model that produces accurate predicted ratings for user–movie pairs and provides the Top-5 recommended movies for each user.
2. **Analytical Objective:** Understand user behavior, genre preferences, and popularity dynamics that drive engagement.
3. **Operational Objective:** Deploy a user-facing app (Streamlit) that returns recommendations in real time.
4. **Business Impact Objective:** Estimate the cost of operation and demonstrate ROI potential via improved engagement metrics (expected higher watch time, lower churn).

5. Data Sources

Our movie data was sourced from the [MovieLens](#) dataset from the GroupLens research lab at the University of Minnesota.

6. Constraints

- Lack of demographic data limits **personalization depth** (e.g., age, location, gender).
- Some movies have **sparse or generic tags**, reducing descriptive richness.
- Ratings are **subjective** and can contain biases.
- No real-time streaming data, so the system relies on static historical records.
- Deployment is limited to a **prototype Streamlit app** (no large-scale production integration).

7. Data Understanding

Dataset Overview

The dataset is structured into four primary files: ratings.csv, movies.csv, tags.csv, and links.csv. Each file contributes unique attributes that, when combined, form a complete picture of user behavior and movie characteristics.

1. ratings.csv

This dataset contains detailed information about how users interact with movies through ratings. It includes the following columns:

- `userId` — Unique identifier for each user.
- `movieId` — Unique identifier for each movie.
- `rating` — Numeric value representing the rating a user assigned to a movie.
- `timestamp` — Records the exact time when the rating was made.

This dataset forms the foundation of the recommendation system by capturing explicit user feedback on various movies over time.

2. movies.csv

This dataset provides descriptive details about the movies themselves. It includes the following columns:

- `movieId` — Unique identifier for each movie.
- `title` — The official movie title, often including the release year.
- `genres` — A pipe-separated list of genres associated with the movie (e.g., Action|Adventure|Comedy).

This structured information allows for detailed understanding and categorization of films based on their content and type.

3. tags.csv

This dataset captures additional descriptive labels users assign to movies. It includes the following columns:

- `userId` — Identifier of the user who created the tag.
- `movieId` — Identifier of the tagged movie.
- `tag` — Free-text label or phrase associated with the movie.
- `timestamp` — Time when the tag was created.

These tags provide an extra layer of semantic meaning that reflects how users perceive and describe different films.

4. links.csv

This dataset acts as a bridge between the MovieLens dataset and external movie databases. It includes the following columns:

- `movieId` — Unique identifier used within MovieLens.
- `imdbId` — Identifier corresponding to the IMDb database.
- `tmdbId` — Identifier corresponding to The Movie Database (TMDb).

By linking these IDs, it becomes possible to integrate additional external data sources such as budgets, cast, crew, and reviews, enriching the overall dataset.

8. Data Preparation

Preparing the dataset transforms it from raw, fragmented files into a clean, structured, and feature-enriched dataset suitable for EDA and modeling. Each step was carefully executed to ensure accuracy and analytical readiness.

1. Data Import and Inspection

- Imported all four datasets into the analysis environment.
- Conducted an initial inspection to confirm column structure, encoding, and row counts.
- Verified column data types to identify required transformations.
- Checked and confirmed that rating values stayed within the valid range (0.5–5.0).

2. Dataset Integration

- Merged links.csv with movies.csv on movieId to create links_movies dataset.
- Linked links_movies with ratings.csv to create final_df.

3. Feature Engineering

- Extracted release year from movie titles for trend analysis.

4. Handling Missing Values

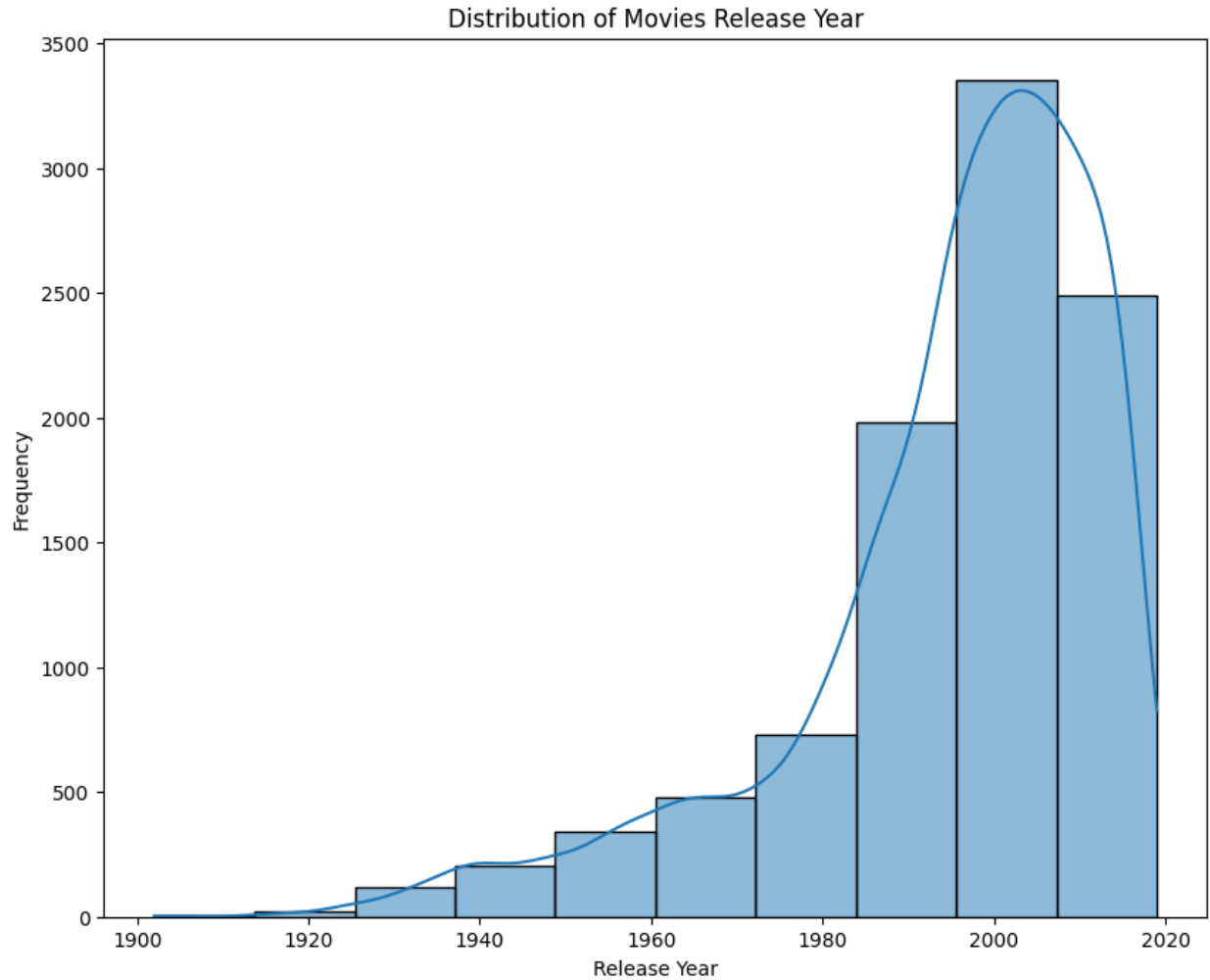
- Checked for null values in our merged dataset.
- Imputed years where release years were missing after splitting the year from the movie title.

Final DataFrame Overview

- Total rows: 100,836
- Unique users: 610
- Unique movies: 9742
- Duplicates removed: 0
- Missing after cleaning: 0
- Columns: userId, movieId, title, year, genres, rating.

9. Exploratory Data Analysis (EDA)

Univariate Analysis

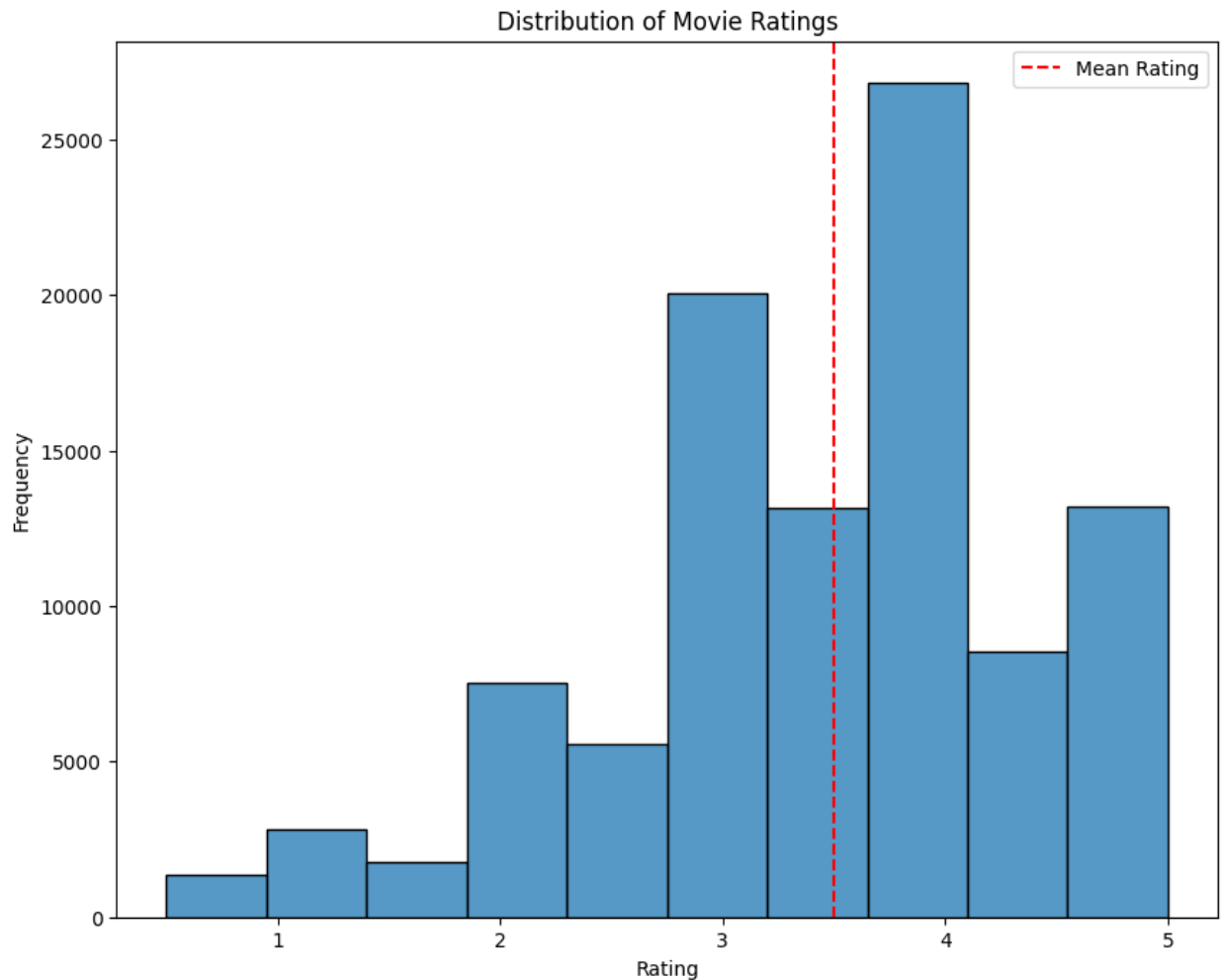


Observations

- Most of the movies that were reviewed were released after early 1980s.
- The oldest movie in our dataset was released around the 1900s and the latest movie added into our dataset seems to be around 2018.

Insight

- Most movies in the dataset are concentrated in the **late 1990s to mid-2000s**, with a noticeable peak around **1995–2005**. This reflects the time period when the MovieLens platform collected the bulk of its data, indicating that most recommendations will be driven by titles from this era.



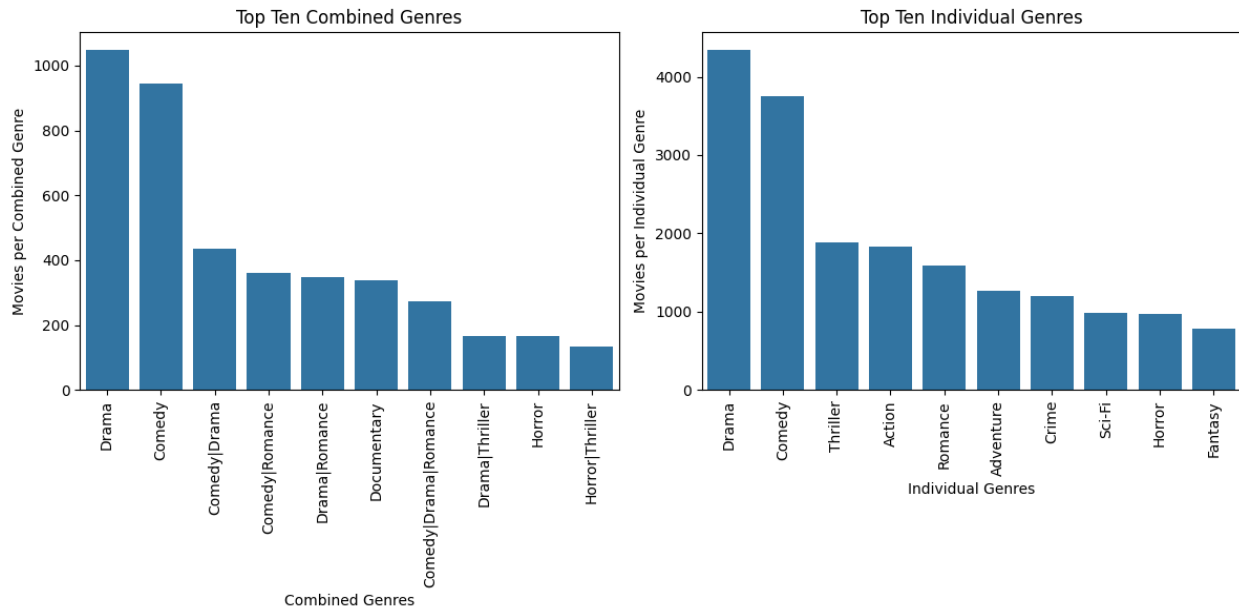
Observations

- Over 25,000 movies were given a rating of 4.
- Less than 5,000 movies were given a rating of less than 1.5
- The mean rating of our dataset was around 3.5

Insight

- Ratings are heavily skewed toward higher values, especially between 3.0 and 5.0. This reveals a positive bias in user feedback, where users tend to rate movies they like more often than those they dislike. This pattern is important for modeling because it can lead to higher predicted ratings on average.

Genre Distribution



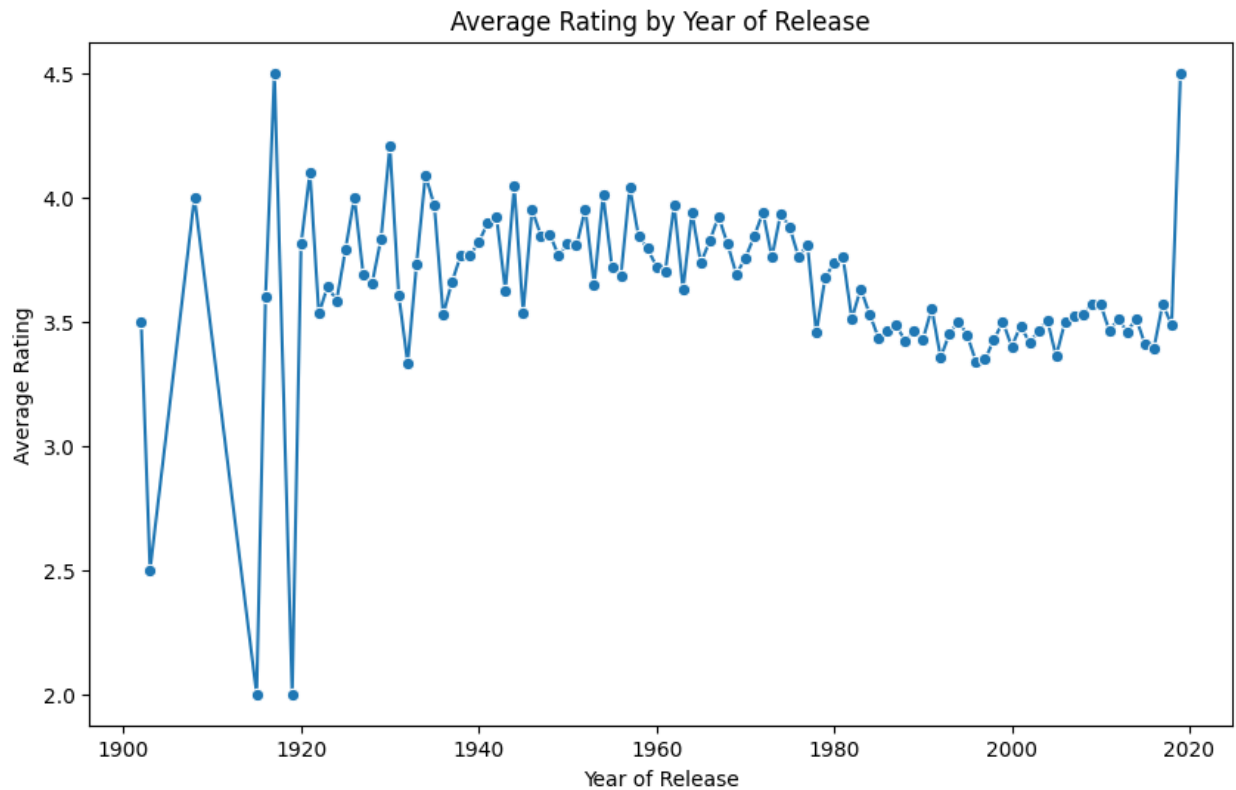
Observations:

- Drama and comedy are the most dominant genres individually and also when combined with other genres.
- Drama is also the most present genre in the other combinations followed by comedy.
- A combination of comedy and drama genres are third highest in the combined genres plot.

Insight:

- A few genres dominate the dataset, particularly Drama, Comedy, and Action. These categories appear frequently across movies, indicating they have a strong influence on user engagement and recommendation behavior. Less common genres may require additional handling to avoid under-representation in recommendations.

Bivariate Analysis

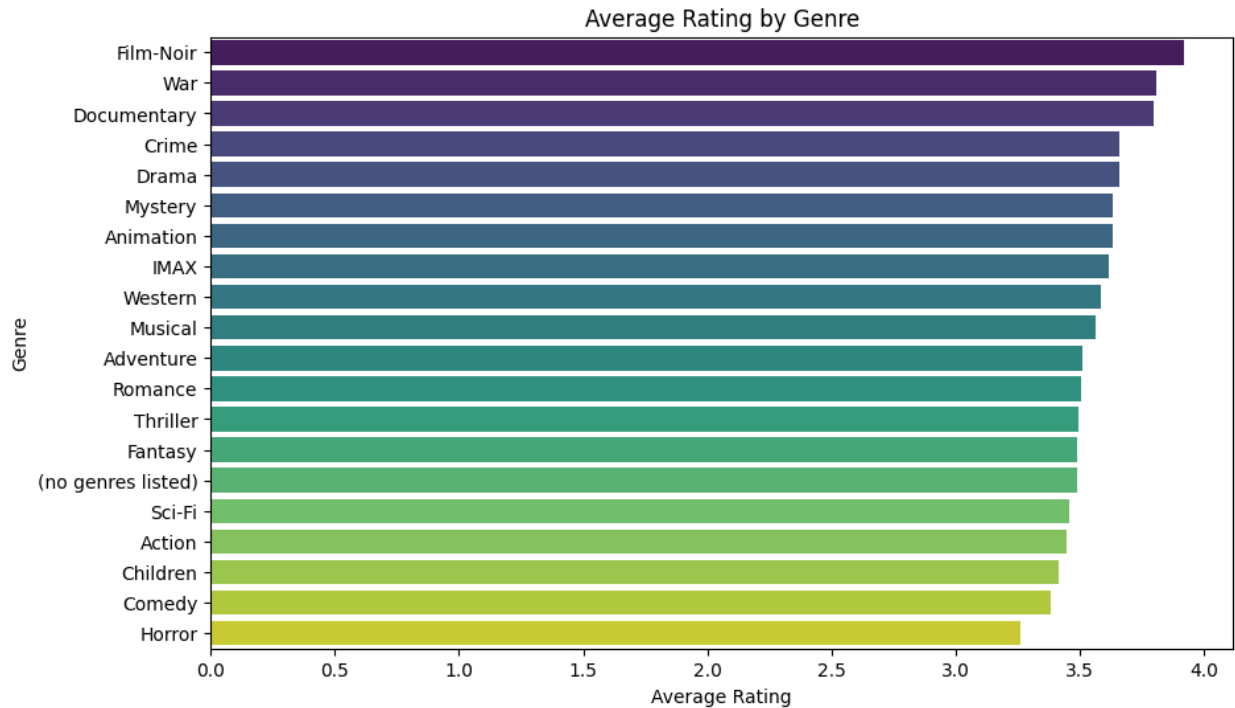


Observations:

- The lowest rating per year was released around the late 1910s
- Most movies released between 1940 and 1980 have an average rating of 3.75 stars.
- There seems to be a drop in rating after 1980 where most movies had an average rating of 3.5 stars. This may be as a result of lack of creativity and repetitive storylines.
- There is a sharp hike in 2020 to an average rating of 4.5. This may be due to technological advancements in the film industry.

Insight:

- There's a slight downward fluctuation in average ratings for more recent years compared to older releases.
- Older movies (especially late 1990s–early 2000s) have consistently higher average ratings, possibly due to lasting classics or nostalgia bias.
- Newer movies may have more mixed or polarized reception, reflecting changing audience preferences over time.



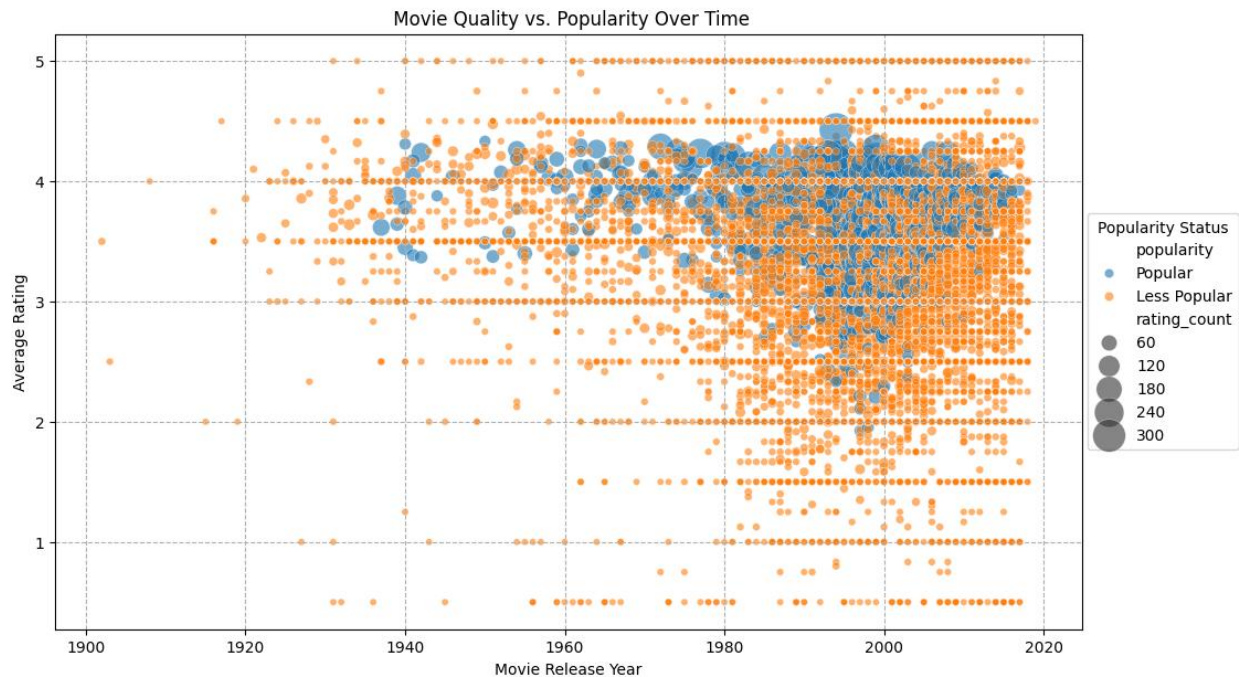
Observations:

- Film Noir movies are black and white and majorly with no sound. They are rated the highest as they were very easy to follow and understand. It had an average of 3.9 stars.
- War, Documentary and Crime come 2nd, 3rd and 4th respectively to show that non fictional films are rated highly.
- Horror was rated the lowest with around 3.2 stars
- There is a very small deviation between the top and bottom rated genres

Insights

- Top-rated genres — *Documentary*, *War*, and *Film-Noir* score the highest, reflecting strong critical appreciation and niche audience engagement.
- Moderately rated genres — *Drama*, *Mystery*, and *Music* show steady appeal and balanced user reception.
- Lower-rated genres — *Horror*, *Animation*, and *Sci-Fi* have more mixed ratings, indicating polarized audience opinions.
- High ratings don't always align with popularity, highlighting the difference between critical quality and mass appeal.
- This pattern suggests tailoring recommendation strategies — prioritizing high-rated genres for quality-focused users and popular genres for entertainment-focused users.

Multivariate Analysis

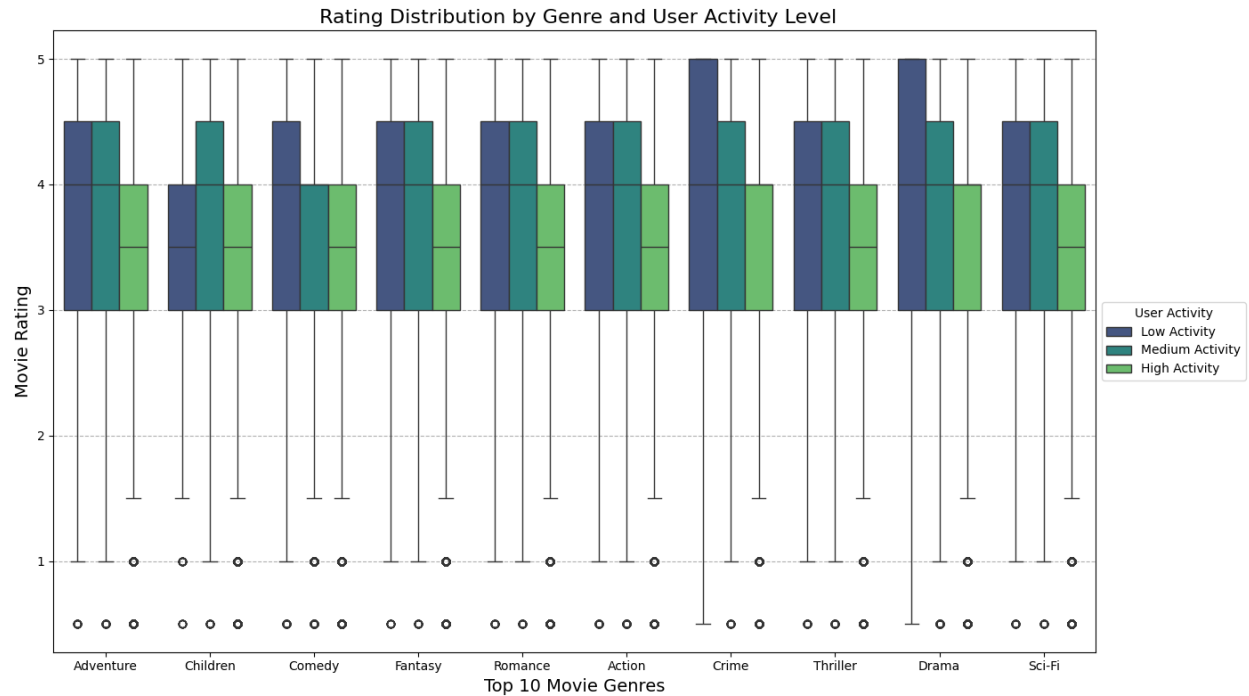


Observation

- Recent decades show a much higher concentration of bubbles showing modern movies receive more ratings
- Very Popular movies tend to cluster in later years and often have higher average ratings.
- Movies released before 1980 have a low rating count
- A few older films appear as highly-rated & highly popular.

Insight:

- Movies with larger budgets tend to have higher revenues and popularity scores, showing a clear commercial pattern.
- A cluster of mid-range budget films shows decent performance but limited popularity spikes, suggesting mainstream but not blockbuster performance.
- A few low-budget films with high popularity indicate successful outliers, possibly cult classics or viral hits.



Observation

- Low Activity (Dark Blue) Easiest Grader consistently gives the highest median ratings across all genres
- High Activity (Green) Strictest Grader consistently gives lower median ratings, especially for "serious" content like Drama and Crime.

Insight:

- Action and Drama movies have the widest distribution of popularity scores, meaning they reach diverse audiences.
- Genres like Documentary and Horror have narrower popularity ranges but can have high ratings, indicating strong but niche fan bases.
- This multivariate view reveals genre as a moderating factor between ratings and popularity.

10. Modeling & Recommendation System

A collaborative filtering model was built using user–item interaction data. The model identifies similar users or movies based on rating patterns and recommends new movies accordingly.

- User and movie IDs were encoded in the Artificial Neural Network (ANN)
- Predictions were generated using kNNBasic, SVD and NMF.
- The model was evaluated based on their RMSE and MAE scores. The lower the scores, the better the model.

Modeling Summary

We evaluated recommenders on cross-validation.

Model	Root Mean Squared Error (RMSE)	Mean Absolute Error (MAE)
Singular Value Decomposition	0.87	0.67
Singular Value Decomposition (Hyper-tuned)	0.85	0.65
k-Nearest Neighbor Basic	0.97	0.74
k-Nearest Neighbor Basic(Hyper-tuned)	0.97	0.74
Non-Negative Matrix Factorization	0.92	0.70
Artificial Neural Network	2.71	2.51

- SVD with hyper-parameter tuning was selected as it gave us the lowest RMSE and MAE.

Deployment Plan using Streamlit

1. Save trained model using the pickle library into Model folder
2. Connect the model and design a simple user interface

Requirements:

- Python 3.11
 - Pandas,
 - Numpy,
 - Scikit-surprise,
 - Scikit-learn,
 - Streamlit
3. On the command line run this script Streamlit: run recommender_system_app.py.

Limitations & Mitigations

- **Cold-start (new users/items)** → onboarding quiz; popularity-weighted fallback; blend content signals.
- **Sparse data** → periodic retraining; incremental updates.

11. Deployment

The recommendation system was deployed using **Streamlit**, a lightweight Python framework for building interactive data apps.

- The model was integrated into a clean user interface.
- Users can enter or select preferences and receive personalized movie recommendations.
- The system runs in real time, fetching top recommended movies dynamically.

This deployment ensures the system is not just a research notebook but a functional, usable Minimum Viable Product

12. Conclusion.

The project successfully developed and deployed a Movie Recommendation System capable of generating relevant suggestions based on user ratings. The analysis confirmed strong user–genre relationships, validating collaborative filtering as an effective method.

The deployment step demonstrated the system’s potential for practical application in real entertainment platforms.

13. Recommendations.

- Integrate more real-time data sources to enhance model adaptability.
- Incorporate hybrid models (content + collaborative) for improved accuracy.
- Add user feedback mechanisms to continuously retrain and improve recommendations.
- Deploy on a cloud platform to support more users concurrently.
- Include visual dashboards for insights on user engagement.